ELSEVIER

Diabetes Research and Clinical Practice 28 (1995) 103–117

DIABETES RESEARCH AND CLINICAL PRACTICE

Intensive insulin therapy prevents the progression of diabetic microvascular complications in Japanese patients with non-insulin-dependent diabetes mellitus: a randomized prospective 6-year study

Yasuo Ohkubo*[a], Hideki Kishikawa[a], Eiichi Araki[a], Takao Miyata[a], Satoshi Isami[a], Sadatoshi Motoyoshi[a], Yujiro Kojima[b], Naohiko Furuyoshi[b], Motoaki Shichiri[a]

[a]*Department of Metabolic Medicine, Kumamoto University School of Medicine, 1-1-1 Honjo, Kumamoto 860, Japan*
[b]*Department of Ophthalmology, Kumamoto University School of Medicine, 1-1-1 Honjo, Kumamoto 860, Japan*

Received 13 January 1995; revision received 20 April 1995; accepted 24 April 1995

Abstract

To examine whether intensive glycemic control could decrease the frequency or severity of diabetic microvascular complications, we performed a prospective study of Japanese patients with non-insulin-dependent diabetes mellitus (NIDDM) treated with multiple insulin injection treatment. A total of 110 patients with NIDDM was randomly assigned to multiple insulin injection treatment group (MIT group) or to conventional insulin injection treatment group (CIT group). Fifty-five NIDDM patients who showed no retinopathy and urinary albumin excretions <30 mg/24 h at the baseline were evaluated in the primary-prevention cohort, and the other 55 NIDDM patients who showed simple retinopathy and urinary albumin excretions <300 mg/24 h were evaluated in the secondary-intervention cohort. The appearance and the progression of retinopathy, nephropathy and neuropathy were evaluated every 6 months over a 6-year period. The worsening of complications in this study was defined as an increase of 2 or more steps in the 19 stages of the modified ETDRS interim scale for retinopathy and an increase of one or more steps in 3 stages (normoalbuminuria, microalbuminuria and albuminuria) for nephropathy. The cumulative percentages of the development and the progression in retinopathy after 6 years were 7.7% for the MIT group and 32.0% for the CIT group in the primary-prevention cohort ($P = 0.039$), and 19.2% for MIT group and 44.0% for CIT group in the secondary-intervention cohort ($P = 0.049$). The cumulative percentages of the development and the progression in nephropathy after 6 years were 7.7% for the MIT group and 28.0% for the CIT group in the primary-prevention cohort ($P = 0.032$), and 11.5% and 32.0%, respectively, for the MIT and CIT groups in the secondary-intervention cohort ($P = 0.044$). In neurological tests after 6 years, MIT group showed significant improvement in the nerve conduction velocities, while the CIT group showed significant deterioration in the median nerve conduction velocities and vibration threshold. Although both postural hypotension and the coefficient of variation of R-R interval tended to improve in the MIT group, they deteriorated in the CIT group. In conclusion, intensive glycemic control by multiple insulin injection therapy can delay the onset and the progression of diabetic retinopathy, nephropathy and neuropathy in Japanese patients with NIDDM. From this study, the glycemic threshold to prevent the onset and the progression of diabetic microangiopathy is indicated as follows; $HbA_1c < 6.5\%$, FBG < 110 mg/dl, and 2-h post-prandial blood glucose concentration < 180 mg/dl.

* Corresponding author.

0168-8227/95/$09.50 © 1995 Elsevier Science Ireland Ltd. All rights reserved
SSDI 0168-8227(95)01064-K

Keywords: Non-insulin-dependent diabetes mellitus; Microvascular complications; Intensive glycemic control; Multiple insulin injection treatment; Glycemic threshold

1. Introduction

After the discovery and clinical application of insulin, death due to acute diabetic complications such as coma has dramatically decreased. However, diabetic microvascular complications such as retinopathy, nephropathy and neuropathy have now been major problems in diabetic patients. The results of the Diabetes Control and Complications Trial (DCCT) in the U.S. and Canada [1,2], as well as those of several other European and American studies, such as the Steno studies [3,4], the Kroc study [5], the Oslo studies [6,7], the Oxford study [8] and the Stockholm studies [9,10], have indicated that intensive insulin therapy could retard the onset and the progression of nephropathy, neuropathy and retinopathy in patients with insulin-dependent diabetes mellitus (IDDM). According to the results of these studies, the significance of intensive glycemic control in IDDM has now been firmly established. However, the significance of intensive insulin therapy in patients with non-insulin-dependent diabetes mellitus (NIDDM) has not been well evaluated.

In Japan, non-obese, insulin-requiring NIDDM patients have been treated with 1 or 2 daily injections of intermediate-acting insulin. However, this conventional insulin injection therapy may be inappropriate for the prevention or intervention of diabetic complications, since conventional insulin injection therapy could not normalize the postprandial hyperglycemia [11]. Therefore, feasibility of the intensive insulin therapy on the prevention of the onset and the progression of diabetic microvascular complications must be evaluated in NIDDM patients. To address such questions, we designed the Kumamoto study.

The Kumamoto study was a randomized clinical trial, designed to compare intensive insulin therapy, using the multiple insulin injection therapy with the conventional insulin injection therapy, to evaluate their effects on the development and the progression of the microvascular complications in NIDDM patients in both the primary-prevention cohort and the secondary-intervention cohort. The intensive insulin therapy regimen in this study was designed to achieve blood glucose values which were as close to the normal range as possible with the 3 or more daily insulin injections. Two cohorts were designed in order to answer 2 different, but related, questions: Will intensive insulin therapy prevent the development of diabetic microangiopathy in NIDDM patients with no retinopathy and urinary albumin excretion <30 mg/24 h (primary-prevention), and will intensive insulin therapy affect the progression of microangiopathy in NIDDM patients with simple retinopathy and urinary albumin excretion <300 mg/24 h (secondary-intervention)? This report presents the results of 6 years of the Kumamoto study in Japanese patients with NIDDM.

2. Patients and methods

2.1. Study design

NIDDM individuals who had been treated with 1 or 2 daily injections of intermediate-acting insulin in our out-patient clinic were asked to participate in this study. A total of 110 patients was selected on the basis of the following criteria: no retinopathy or simple retinopathy determined by clinical funduscopic evaluation, urinary albumin excretion <300 mg/24 h and serum creatinine level <1.5 mg/dl, absence of diabetic somatic or autonomic neuropathy severe enough to require treatment, <70 years of age, and otherwise healthy (no other findings such as hypertension, hypercholesterolemia, severe diabetic complications or other severe medical conditions). All of the patients were diagnosed as being affected with NIDDM by their characteristics of no history of ketoacidosis, negative islet cell antibody, and daily urinary C-peptide excretion more than 20 μg [12]. Informed consent was obtained from each patient.

One hundred and ten patients were divided into 2 cohorts — the primary-prevention cohort ($n = 55$) and the secondary-intervention cohort ($n = 55$). To be included in the primary-prevention

cohort, the patients were required to have no retinopathy and urinary albumin excretion <30 mg/24 h. To be included in the secondary-intervention cohort, the patients were required to have simple retinopathy and urinary albumin excretion <300 mg/24 h. In addition, these patients were randomly assigned to either conventional insulin injection therapy group (CIT group, $n = 55$) or multiple insulin injection therapy group (MIT group, $n = 55$). The CIT group was administered 1 or 2 daily injections of intermediate-acting insulin. The MIT group was defined as the group that was administered insulin 3 or more times daily (rapid-acting insulin at each meal and intermediate-acting insulin at bedtime). The dosage was adjusted according to the self-monitored results of blood glucose.

The clinical goal of the CIT group was to show no symptoms of hyperglycemia or hypoglycemia, and the glycemic control as close to the fasting blood glucose concentration of <140 mg/dl. The goal of the MIT group was to maintain the blood glucose control as close to the fasting blood glucose concentration (<140 mg/dl), 2-h postprandial blood glucose concentration <200 mg/dl, hemoglobin A_1c (HbA_1c) <7.0% and mean amplitude of glycemic excursions (MAGE) <100 mg/dl.

The baseline characteristics of the study cohorts are shown in Table 1. After 6 years, 102 patients remained in the study, 3 patients died (2 in the MIT group and 1 in the CIT group), 3 patients had moved to another city (1 in the MIT group and 2 in the CIT group), and 2 patients had changed from conventional insulin injection therapy to multiple insulin injection therapy. There was no significant difference at entry in either the primary-prevention cohort or the secondary-

Table 1
Characteristics of patients at the baseline

	Primary-prevention cohort		Secondary-intervention cohort	
	CIT group	MIT group	CIT group	MIT group
Numbers (male/female)	25(12/13)	26(14/12)	25(11/14)	26(12/14)
Age (years)	49 ± 14	47 ± 9	52 ± 15	49 ± 13
Duration of diabetes (years)	6.7 ± 5.3	6.2 ± 5.0	10.3 ± 4.9	10.2 ± 4.2
Body mass index (kg/m^2)	21.2 ± 2.3	21.7 ± 1.8	19.2 ± 2.8	19.3 ± 1.7
Insulin dosage (U/kg of body weight /24 h)	0.39 ± 0.21	0.38 ± 0.19	0.41 ± 0.22	0.42 ± 0.22
Urinary C-peptide excretion (μg/24 h)	26.5 ± 27.1	22.1 ± 23.5	21.7 ± 24.9	18.0 ± 22.4
Hemoglobin A_1c (%)	8.9 ± 1.8	9.2 ± 1.8	9.0 ± 1.9	9.4 ± 1.8
Blood pressure (mmHg)				
Systolic	121 ± 14	119 ± 9	123 ± 9	121 ± 11
Diastolic	70 ± 7	69 ± 6	70 ± 7	70 ± 8
Serum cholesterol (mg/dl)	199 ± 38	198 ± 39	202 ± 43	204 ± 41
Serum triglycerides (mg/dl)	95 ± 37	99 ± 58	97 ± 39	102 ± 56
Serum HDL cholesterol (mg/dl)	53 ± 14	50 ± 14	55 ± 23	52 ± 19
No retinopathy	25	26	0	0
Simple retinopathy	0	0	25	26
Urinary albumin excretion (mg/24 h)	13 ± 12	14 ± 13	41 ± 77	45 ± 78
Urinary NAG excretion (U/24 h)	7.1 ± 1.9	7.1 ± 1.8	7.4 ± 2.9	7.3 ± 2.9
Nerve conduction velocity (m/s)				
Median nerve (motor)	52.5 ± 4.5	51.9 ± 4.7	49.9 ± 5.5	49.7 ± 5.7

CIT, conventional insulin injection therapy; MIT, multiple insulin injection therapy.
Values are expressed as mean ± S.D.

intervention cohort between the CIT and MIT groups with regard to sex distribution, age, duration of diabetes, body mass index (BMI), insulin dosage, urinary C-peptide excretion, HbA_1c, blood pressure, serum cholesterol, serum triglyceride, serum HDL cholesterol, degree of retinopathy, urinary albumin excretion, urinary N-acetyl-β-D-glucosaminidase (NAG) excretion and nerve conduction velocity. Follow-up examinations of glycemic control and diabetic microvascular complications were performed at 3, 6, 12, 18, 24, 30, 36, 42, 48, 54, 60, 66 and 72 months after initiation of the study.

2.2. Evaluation of glycemic control

Metabolic control was estimated by HbA_1c, fasting blood glucose (FBG), mean blood glucose (MBG), *M*-value and MAGE from diurnal blood glucose profiles which consisted of 11 tests (fasting, 1 and 2 h post-prandial, bedtime and fasting at the next morning) in 1-day hospitalization periodically. The mean values for HbA_1c, FBG, MBG, *M*-value and MAGE during the study were calculated to evaluate the glycemic control during a period of 3–72 months. Severe hypoglycemia is defined as an event with symptoms consistent with hypoglycemia in which the patients required the assistance of another person, and which was associated with a blood glucose level < 50 mg/dl and prompt recovery after intravenous glucose loading.

2.3. Evaluation of diabetic microvascular complications

Retinopathy. All of the patients had direct ophthalmoscopy, with pupils dilated. Fundoscopic findings were evaluated by at least 2 examiners (an ophthalmologist and an internist) followed by color fundus photography and fluorescein angiography. The degrees of retinopathy were determined by the 2 eye examiners for each patient in accordance with the ETDRS interim scale [13–15], in which a scale of 19 stages is used to represent the overall extent of retinopathy in both eyes (Table 2).

The development and the progression of retinopathy were defined as the change of at least

Table 2
Modified ETDRS grading of diabetic retinopathy

Stage	Grading criteria
1.	No diabetic retinopathy.
2.	The presence of simple diabetic retinopathy (SDR) insufficient to be classified as group 4, the presence of microaneurysms or retinal hemorrhage < standard photograph 2A without macular edema or hard exudates > standard photograph 3 and without macular edema on fluorescein angiography (on 1 eye only).
3.	2 on both eyes.
4.	SDR insufficient to be classified as group 6, specifically, at least 1 microaneurysm or retinal hemorrhage < standard photograph 2A plus macular edema represented by either a thickened retina or hard exudates > standard photograph 3 or with evidence of macular edema on fluorescein angiography (on 1 eye only).
5.	4 on both eyes.
6.	Evidence of any 1 of the following in any 1 of the 2 standard fields (disc and macula): hemorrhages and microaneurysms > standard photograph 2A, soft exudates, venous beading, or intra-retinal microvascular abnormalities (IRMA) (on 1 eye only).
7.	6 on both eyes.
8.	Various combinations of hemorrhages and microaneurysms, soft exudates, venous beading, and IRMA in more than 1 standard field (on 1 eye only).
9.	8 on both eyes.
10.	Flat NVE (neovascularization elsewhere) present NVD (neovascularization on disc or within 1 disc diameter of the disc) absent (on 1 eye only).
11.	10 on both eyes.
12.	Flat NVD present: NVE absent (on 1 eye only).
13.	12 on both eyes.
14.	Flat NVD and NVE present (on 1 eye only).
15.	14 on both eyes.
16.	The same as groups 10–15 with regard to the NVD and NVE, but the neovascularization must be elevated above the surface of retina by 1/4 disc diameter, but not by more than 1 disc diameter (on 1 eye only).
17.	16 on both eyes.
18.	The same as groups 10–15 with regard to NVD and NVE, but the neovascularization must be elevated in at least 1 place by more than 1 disc diameter off retinal plane (on 1 eye only).
19.	18 on both eyes.

Standard photographs: the Early Treatment Diabetic Retinopathy Study (ETDRS) standard photographs.

2 steps from stage 1 in the primary-prevention cohort, and as 2 or more steps change from stages 2–5 in the secondary-intervention cohort. This definition was chosen because it represented a measure of clinically important worsening.

Nephropathy. The patients with nephropathy were divided into 3 stages depending on their urinary albumin excretion — normoalbuminuria (< 30 mg/24 h), microalbuminuria (30–300 mg/24 h) or albuminuria (> 300 mg/24 h) [3,16]. The change from any stage to a higher stage was defined as progression. Whenever the urinary albumin excretion exceeded 30 mg/24 h in patients with normoalbuminuria, the urine was analyzed for the presence of bacteria. If bacteria were present, a subsequent 24-h collection of urine was obtained after treatment for urinary tract infections. On the other hand, to evaluate the renal tubular function, the 24-h urinary NAG excretion was measured every 6 months and compared between the values at entry and those after 6 years.

Neuropathy. Peripheral nerve function was evaluated by temperature-controlled nerve conduction velocity in the median nerve on one side and by vibration threshold on the radial styloid process of the arm and the medial malleolus of the leg on both sides. The measurements of vibration threshold were taken 3 times on both sides and their averages were calculated. These neurological indices at entry and after 6 years of the study in both the CIT and the MIT group were compared in the combined cohort.

The coefficient of variation (C.V.) of R-R interval and the postural hypotension were used to evaluate the indices of the function of parasympathetic and sympathetic nerves, respectively. The C.V. of R-R interval on electrocardiogram was evaluated both during 100 pulses of normal breathing after 5 min of bed rest and during 100 pulses while deep breathing. In the postural hypotension, blood pressure was measured while the patient was in the supine position after 5 min of bed rest and after standing. The decrease in systolic blood pressure caused by standing was determined. These neurological indices at entry and after 6 years of the study in both the CIT and MIT groups were compared in the combined cohort.

2.4. Evaluation of the relationship between the worsening of retinopathy and nephropathy, and glycemic control indices

The relationship between the worsening of retinopathy and nephropathy, and glycemic control indices, expressed as HbA_1c, FBG and 2-h post-prandial blood glucose concentration, was evaluated in the combined cohort. The crude rate within the 6 grades evaluated according to their glycemic control indices was calculated and expressed as 100 patient-years. The regression lines estimated as a function of the log of mean glycemic control indices were calculated.

2.5. Assays

Blood glucose concentration was assayed using the GOD-PAP method (Autoanalyzer ™, Miles Inc., Tarrytown, NY, USA). HbA_1c value was assayed using high performance liquid chromatography (HLC 723 GHb type 2, Tosoh Co. Ltd, Tokyo, Japan, normal range: 4.8 to 6.4%). Fundoscopic findings were recorded using the CR-45 camera (Cannon Co. Ltd., Tokyo, Japan). Urinary albumin excretion was measured by turbidimetric immunoassay (Automatic analyzer 7150 type, Hitachi Ltd., Tokyo, Japan). Urinary NAG excretion was measured by a colorimetric method using *p*-nitrophenyl-acetyl-β-D-glucosaminide substrates (SSR-XT, Miles Inc., Tarrytown, NY, USA). Nerve conduction velocity, vibration threshold and C.V. of R-R interval were measured by the Neuromatic 2000 ™ (Dantec, Skovlunde, Denmark), the Vibratory Sensation Meter (TM-31A, Medic International, Tokyo, Japan) and the Electro Cardiograph Analysis System 12 (Nihon Kohden Co. Ltd., Tokyo, Japan), respectively.

2.6. Statistical analysis

The Wilcoxon rank-sum test was used to compare the 2 treatment groups within the cohort with regard to ordinal and numerical variables, and the chi-square test was used for comparisons of categorical variables. The life-table method was used to estimate the cumulative incidence of events, with adjustment for periodically timed assessments. The difference between 2 cumulative incidence curves was tested by the Mantel (log-rank) test [17]. The percentages of risk reduction

by intensive glycemic control using multiple insulin injection therapy were calculated by subtracting the relative risk of multiple insulin injection therapy as compared with conventional insulin injection therapy from 1 and multiplying by 100. The relative risks of retinopathy and nephropathy were estimated as the crude relative risk in the primary-prevention cohort, the secondary-intervention cohort and the combined cohort. In addition, the relative risks were also calculated using proportional-hazards analysis only in the combined cohort. Event rates of complications were presented as the number of events per 100 patient-years, based on the ratio of observed number of events to the total number of patient-years of exposure. All outcomes were analyzed on the basis of the original treatment assignments. All results were expressed as mean ± S.D. $P < 0.05$ was considered to be statistically significant.

3. Results

3.1. Glycemic control

As shown in Fig. 1, near normoglycemia was obtained in the MIT group during the third month after the initiation of intensive insulin therapy and was maintained over the 6 years of the study, while the glycemic control in the CIT group did not change significantly. The resultant mean values of FBG, HbA_1c, MBG, *M*-value and MAGE over 6 years were significantly ($P < 0.001$) lower in the MIT group than in the CIT group (FBG,

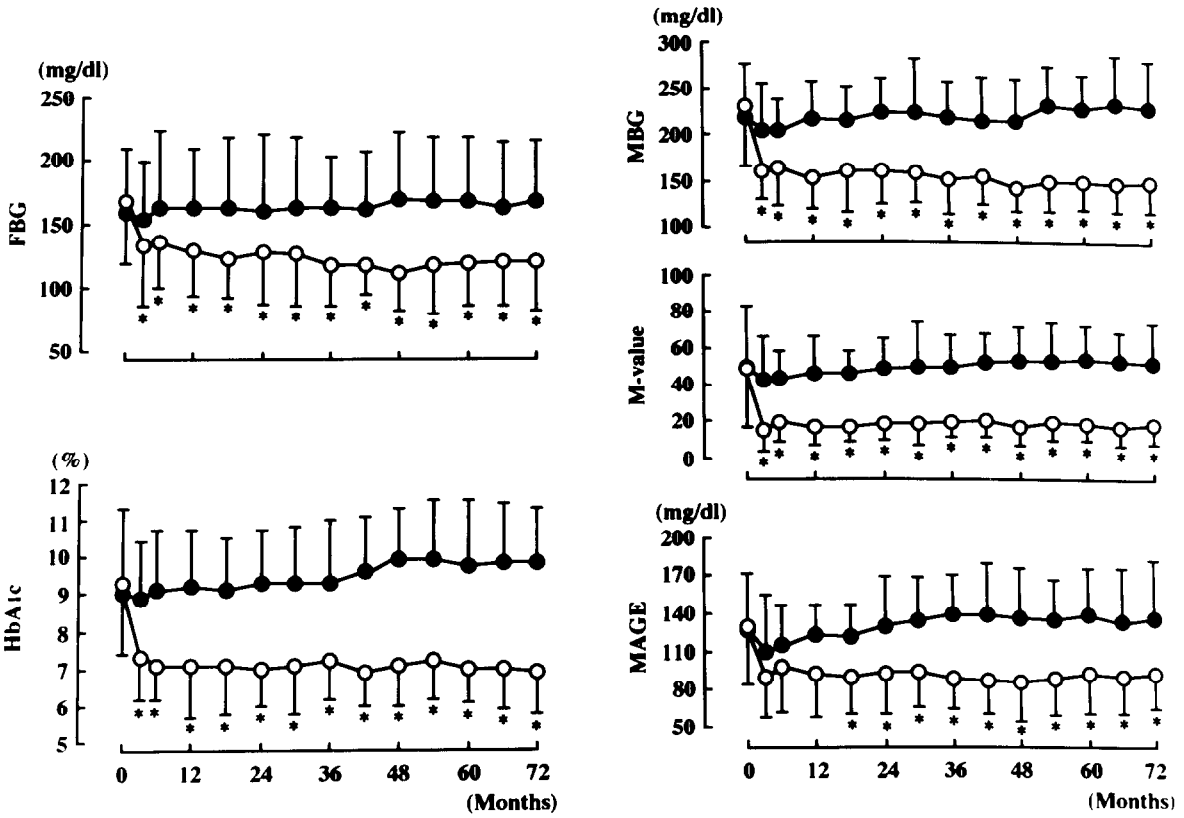


Fig. 1. Glycemic control as assessed by FBG, HbA_1c, MBG, *M*-value and MAGE over a 72-month period in NIDDM patients receiving intensive (○) and conventional (●) insulin injection therapy. Mean values ± S.D. are shown. *$P < 0.05$ vs. conventional insulin injection therapy.

126 ± 36 vs. 164 ± 50 mg/dl; HbA_1c, 7.1 ± 1.1 vs. 9.4 ± 1.5%; MBG, 157 ± 34 vs. 221 ± 45 mg/dl; *M*-value, 18 ± 10 vs. 49 ± 19; MAGE, 91 ± 30 vs. 128 ± 36 mg/dl). On the other hand, there was a slight but not significant decrease in insulin dosage from the baseline to 6 years in the MIT and CIT groups (MIT group, 0.40 ± 0.28 to 0.36 ± 0.35 U/kg of body weight/24 h; CIT group, 0.41 ± 0.28 to 0.44 ± 0.35 U/kg of body weight/24 h).

3.2. Retinopathy

Primary-prevention cohort. The cumulative percentage of patients with the development of retinopathy after 6 years in the MIT group was significantly lower when compared with those in the CIT group (7.7 vs. 32.0%, $P = 0.039$, Fig. 2A). During the 6-year period, the development of retinopathy appeared in 2 patients in the MIT group and 8 patients in the CIT group. No patient of either group required photocoagulation therapy.

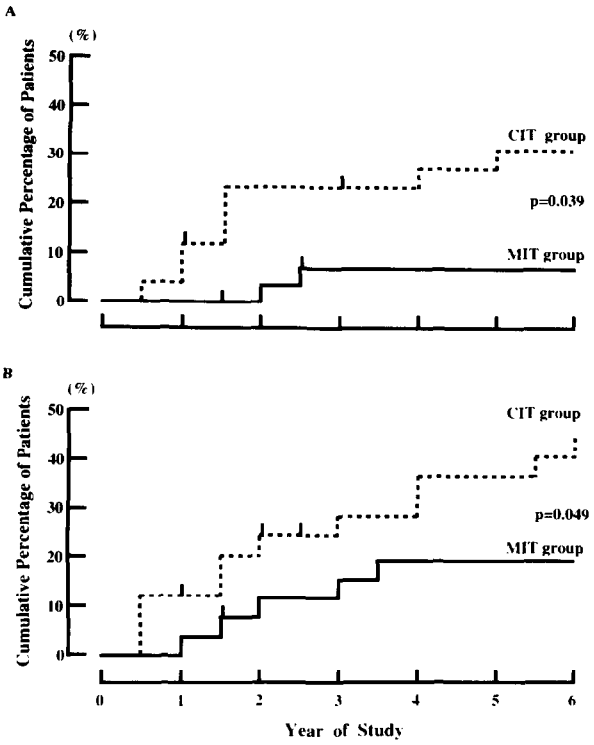


Fig. 2. Cumulative incidences of a change in retinopathy in NIDDM patients treated by intensive (solid line) and conventional (dashed line) insulin injection therapy. A change in the severity of retinopathy was defined as at least 2 steps up from the baseline (ETDRS interim scale). The primary-prevention and secondary-intervention cohorts are shown in panels A and B, respectively. Dropout patients are indicated by a short vertical line on the solid and dashed lines. *P*-values were tested using the Mantel (log-rank) test.

Secondary-intervention cohort. The cumulative percentage of patients with the progression of retinopathy was 19.2% in the MIT group and 44.0% in the CIT group after 6 years ($P = 0.049$, Fig. 2B). During the 6-year period, the progression of retinopathy was found in 5 patients in the MIT group and 11 patients in the CIT group. Two patients in the MIT group and 2 patients in the CIT group were diagnosed as having preproliferative retinopathy, and 1 patient in the CIT group showed proliferative retinopathy during this study. All of them required photocoagulation therapy.

Combined cohort. In the combined cohort consisting of both the primary-prevention cohort and the secondary-intervention cohort, the cumulative percentage of patients with the worsening in retinopathy after 6 years was significantly lower in the MIT group when compared with those in the CIT group (13.4 vs. 38.0%, $P = 0.007$). Intensive glycemic control by multiple insulin injection therapy reduced the average risk of worsening in retinopathy by 69% (95% confidence interval, 24 to 87%) during the entire study period. All of the patients who had serious retinopathy requiring photocoagulation in both the MIT and the CIT groups showed poor glycemic control (the mean value of HbA_1c was 9.6 ± 1.0%).

3.3. Nephropathy

Primary-prevention cohort. The cumulative percentage of patients with the development of nephropathy after 6 years was significantly lower in the MIT group than in the CIT group (7.7 vs. 28.0%, $P = 0.032$, Fig. 3A). During the 6-year period, 5 patients in the CIT group and 2 patients in the MIT group developed microalbuminuria, and 2 patients in the CIT group developed albuminuria.

Secondary-intervention cohort. The cumulative percentage of patients with the progression of

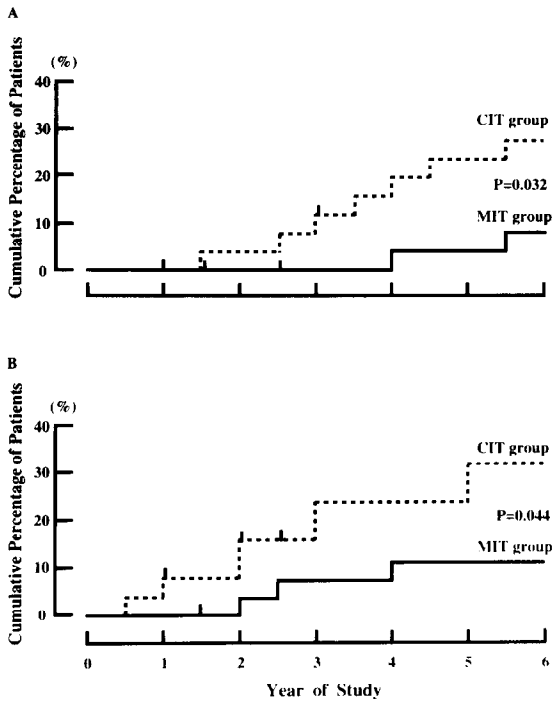


Fig. 3. Cumulative incidences of a change in nephropathy in NIDDM patients treated by intensive (solid line) and conventional (dashed line) insulin injection therapy. A change in the severity of nephropathy was defined as 1 or more stages up among 3 stages (normoalbuminuria, microalbuminuria and albuminuria). The primary-prevention and secondary-intervention cohorts are shown in panels A and B, respectively. Dropout patients are indicated by short vertical lines on the solid and dashed lines. *P*-values were tested using the Mantel (log-rank) test.

nephropathy was significantly lower in the MIT group than in the CIT group (11.5 vs. 32.0%; $P = 0.044$), as shown in Fig. 3B. During the 6-year period, 6 patients in the CIT group and 3 patients in the MIT group developed microalbuminuria, and 2 patients in the CIT group developed albuminuria.

Combined cohort. In the combined cohort, the cumulative percentage of patients with the worsening in nephropathy after 6 years was significantly lower in the MIT group than in the CIT group (9.6 vs. 30.0%, $P = 0.005$). Intensive glycemic control by multiple insulin injection therapy reduced the average risk of worsening in nephropathy by 70% (95% C.I., 14–89%) during the entire study period. On the other hand, the mean urinary NAG excretion in the CIT group increased from 7.3 ± 2.5 U/24 h at the beginning of the study to 7.7 ± 4.2 U/24 h after 6 years, but decreased in the MIT group from 7.2 ± 2.4 U/24 h at entry to 6.2 ± 2.1 U/24 h after 6 years. After 6 years, the difference between the 2 groups was significant ($P < 0.05$).

3.4. Neuropathy

Results of nerve conduction velocity, vibration threshold, postural hypotension and C.V. of R-R interval in the combined cohort at entry and after 6-year period are shown in Table 3.

The median nerve conduction velocities (both motor and sensory nerves) significantly increased after 6 years in the MIT group as compared with the values at entry, while the median sensory nerve conduction velocity significantly decreased in the CIT group. There were significant differences in the motor and sensory nerve conduction velocities between both groups after 6 years of the study ($P < 0.05$). The vibration thresholds in the MIT group showed a slight, but not significant increase after 6 years, while those in the CIT group significantly ($P < 0.05$) increased after 6 years. There were significant ($P < 0.05$) differences in arm vibration thresholds between the MIT and the CIT groups after 6 years of the study.

The MIT group showed only slight improvements in the postural hypotension and the C.V. of R-R interval in both at rest and during deep breathing, while the CIT group showed slight deterioration in both tests.

3.5. Evaluation of the relation between the worsening of retinopathy and nephropathy, and glycemic control indices

Comparison of glycemic control indices, with or without worsening of retinopathy and nephropathy, are shown in Table 4. The patients without worsening of microangiopathy over the study period exhibited significantly ($P < 0.001$) lower mean glycemic control indices than the patients with worsening. We analyzed the relationship between the rate of worsening of retinopathy and nephropathy, and glycemic indices expressed as HbA_1c, FBG and 2-h post-prandial blood glucose

Table 3
Peripheral nerve function and autonomic nerve function in CIT and MIT groups at entry to the study and after 6 years in the combined cohort

	CIT group		MIT group	
	At entry	After 6 years	At entry	After 6 years
Peripheral neuropathy				
Nerve conduction velocity (m/s)				
Median nerve (motor)	51.1 ± 6.3	50.2 ± 5.6	50.8 ± 6.4	53.2 ± 7.1*,**
Median nerve (sensory)	56.4 ± 6.4	54.7 ± 7.0	56.0 ± 5.7	58.2 ± 6.4*,**
Vibration threshold (μm)				
Arm	10.7 ± 4.9	15.0 ± 10.4	10.4 ± 5.0	11.4 ± 7.1*
Leg	22.6 ± 12.6	37.2 ± 15.9**	21.1 ± 12.9	23.8 ± 12.9
Autonomic neuropathy				
Postural hypotension (ΔBp) (mmHg)	8.7 ± 8.5	9.6 ± 8.4	8.3 ± 10.0	6.7 ± 1.0
C.V. of R-R interval (%)				
At rest	2.5 ± 1.1	2.4 ± 1.2	2.5 ± 0.8	2.7 ± 1.0
During deep breathing	5.2 ± 1.9	4.9 ± 1.7	5.3 ± 1.5	5.7 ± 1.6

CIT, conventional insulin injection therapy; MIT: multiple insulin injection therapy.
Values are expressed as mean ± S.D.
*$P < 0.05$ vs. CIT group.
**$P < 0.05$ vs. at entry.

concentration in the combined cohort (Fig. 4). These analyses showed a continuously increasing risk of the worsening of retinopathy and nephropathy with increasing HbA_1c, FBG and 2-h post-prandial blood glucose concentration. No worsening of retinopathy and nephropathy was observed in the NIDDM patients whose HbA_1c, FBG and 2-h post-prandial blood glucose concentration were below 6.5%, 110 mg/dl and 180 mg/dl, respectively.

Table 4
Relationship between the worsening[a] of microvascular complications and glycemic control indices

Group	HbA_1c (%)	FBG (mg/dl)	2-h post-prandial blood glucose concentration (mg/dl)
Retinopathy			
Stationary	7.4 ± 1.4*	134 ± 28*	207 ± 80*
Worsening	9.3 ± 1.1	163 ± 28	255 ± 25
Nephropathy			
Stationary	7.7 ± 1.4*	136 ± 28*	205 ± 39*
Worsening	9.6 ± 1.2	167 ± 27	253 ± 25

[a]Worsening was defined as the case who displayed the development and progression of retinopathy and nephropathy.
Values are expressed as mean ± S.D.
*$P < 0.001$ vs. worsening group.

3.6. Hypoglycemia

Over the entire study period, 6 patients in the MIT group and 4 patients in the CIT group had 1 or more episodes of mild hypoglycemic reactions, but no patient experienced episodes of coma, seizure, or severe hypoglycemia that required the assistance of another person.

4. Discussion

The fact that the microvascular complications associated with IDDM can be prevented or delayed by intensive insulin therapy [1–10], has provided us with tremendous hope for the future management of patients with diabetes. One of the important questions raised by these studies evaluated in IDDM is to what extent their results

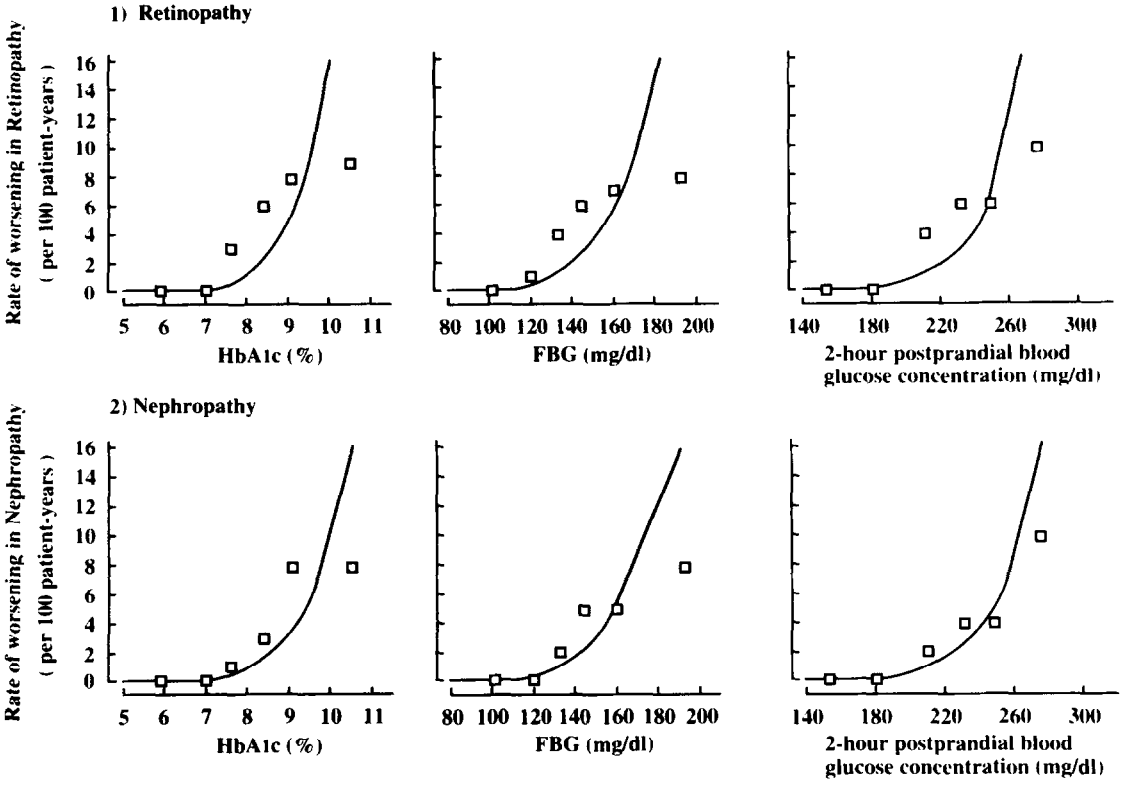


Fig. 4. Relationship between the rate of worsening of retinopathy and nephropathy, and glycemic control indices expressed as HbA_1c, FBG and 2-h post-prandial blood glucose concentration in the combined cohort. Square indicates the crude rate within the 6 grades evaluated according to their glycemic control levels; each square corresponds to more than 100 patient-years. The solid lines are regression lines estimated as a function of the log of the mean HbA_1c, FBG and 2-h post-prandial glucose concentration.

can or should be extrapolated to the management of patients with NIDDM. To address such a question, we studied the effect of intensive glycemic control on diabetic microvascular complications in Japanese patients with NIDDM.

In this study, we demonstrated that intensive glycemic control with multiple insulin injection therapy effectively delayed the onset and the progression of diabetic retinopathy, nephropathy and neuropathy in Japanese NIDDM patients. Table 5 demonstrates the reduction in risk for the development and the progression of diabetic microvascular complications in the Kumamoto study for NIDDM in comparison with that in DCCT for IDDM. Although different statistical methods were applied in the Kumamoto study and DCCT because of the different number of patients studied, the results of reduction in risk with intensive insulin therapy in the Kumamoto study for NIDDM patients were almost comparable to those in DCCT for IDDM patients.

In the evaluation of retinal findings, we used the ETDRS interim scale of 19 stages, different from DCCT in which a scale of 25 stages was used [2]. This classification system allowed us to evaluate the retinal findings systematically and effectively. The cumulative percentage of patients showing the worsening in retinal findings was significantly lower in the MIT group than in the CIT group. With intensive insulin therapy, the risks of retinopathy calculated by crude relative risk were reduced by 76, 56 and 65% in the primary-

Table 5
Development and progression of microvascular complications of diabetes and reduction in risk with intensive insulin therapy in the Kumamoto study and the DCCT

Complications	Primary-prevention cohort			Secondary-intervention cohort			Combined cohort	
	C.T.[a] Event rate[a]	I.T.[a] Event rate[a]	R.R.[b] (%)	C.T.[a] Event rate[a]	I.T.[a] Event rate[a]	R.R.[b] (%)	R.R.[b] (%)	
Kumamoto Study in patients with NIDDM								
≥2-Step sustained retinopathy (scale of 19 stages)	5.3	1.3	76*	7.3	3.2	56*	65*	(69)**
Severe non-proliferative or proliferative retinopathy	–	–	–	2.0	1.2	40*	40*	
Photocoagulation	–	–	–	2.0	1.2	40*	40*	
Urinary albumin excretion (mg/24 h)								
≥30	3.3	1.3	62*	4.0	1.9	52*	57*	(70)**
>300	1.3	0	100*	1.3	0	100*	100*	
DCCT in patients with IDDM								
≥3-Step sustained retinopathy (scale of 25 stages)	4.7	1.2	76**	7.8	3.7	54**	63**	
Severe non-proliferative or proliferative retinopathy	–	–	–	2.4	1.1	47**	47**	
Photocoagulation	–	–	–	2.3	0.9	56**	51**	
Urinary albumin excretion (mg/24 h)								
≥40	3.4	2.2	34**	5.7	3.6	43**	39**	
≥300	0.3	0.2	44**	1.4	0.6	56**	54**	

CT., conventional therapy; I.T., intensive therapy., R.R., risk reduction.
[a]Defined as rates of development and progression of complications per 100 patient-years.
[b]Represents the comparison of intensive with conventional insulin injection therapy.
*Expressed as a percentage and calculated using crude relative risk analysis in the Kumamoto study.
**Expressed as a percentage and calculated using proportional-hazards analysis in the Kumamoto study and DCCT.

prevention, the secondary-intervention and the combined cohorts. The risk reduction calculated by proportional-hazard analysis in the combined cohort in the Kumamoto study was similar to that in DCCT (69 vs. 63%, respectively). Intensive insulin therapy reduced the risk of severe nonproliferative or proliferative retinopathy by 40% and that of treatment of photocoagulation by 40%. From these results, we confirmed that the near-normal glycemic control was associated with the delay in the development and the progression of diabetic retinopathy. In previous studies of IDDM patients [2,5,18,19], the appearance of soft exudates or intra-retinal microvascular abnormalities has been reported after the initiation of intensive insulin therapy. Initial worsening has been reported to occur during the first year after the start of the intensive insulin therapy. In our study, however, only 1 patient in the MIT group showed the appearance of small bleeding within 6 months after the initiation of intensive therapy. This small bleeding disappeared over the course of the continued intensive insulin therapy.

Intensive glycemic control was also effective for diabetic nephropathy. Intensive insulin therapy reduced the risk of microalbuminuria by 62% and albuminuria by 100% in the primary-prevention cohort. It also reduced the risk of microalbuminuria by 52% and albuminuria by 100% in the secondary-intervention cohort, respectively. The risk reduction calculated by proportional-hazard risk analysis was more in the Kumamoto study for NIDDM than that in DCCT (risk reduction in the combined cohort, 70 vs. 39–54%, respectively). A striking difference in the risk reduction in nephropathy was observed between the Kumamoto study for NIDDM and DCCT for IDDM. The reason why the intensive glycemic control significantly reduced the risk of progression of microalbuminuria or albuminuria in the Kumamoto study has not been explained thoroughly, but might possibly be explained by the different susceptibility to nephropathy between the types of diabetes, NIDDM and IDDM, and between the races, Caucasian and Asian. It has been known that Japanese NIDDM patients are susceptible to nephropathy [20,21]. Further studies on the multicenter randomized prospective study with many Japanese NIDDM patients should, anyhow, be necessary to solve this question. The decrease in the 24-h urinary NAG excretion in the MIT group also supported a beneficial effect of intensive glycemic control on renal tubular function. In previous studies [22–24], high blood pressure has been reported to be responsible for the impairment of the renal function. In our study, blood pressure in both groups did not significantly change over the study period and there were no significant differences in mean blood pressure between those with and without worsening in nephropathy (data not shown). Longer follow-up studies might be needed to elucidate the effect of blood pressure on the progression of nephropathy.

After the 6-year period, there were significant differences between the CIT and MIT groups in the nerve conduction velocities and arm vibration threshold. The postural hypotension and C.V. of R-R interval in the MIT group tended to improve after 6 years, while these indices in the CIT group tended to deteriorate. Therefore, we concluded that intensive glycemic control would have a beneficial effect on both somatic and autonomic nerve function. Our Kumamoto study suggested that somatic nerve function seemed to improve more easily than autonomic nerve function by intensive insulin therapy. This finding was consistent with the findings that reversible changes were possible only in nerve fibers that were myelinated [25,26].

Since DCCT reported that the relative youth of the patients made the detection of treatment-related differences in rates of macrovascular events unlikely [2], the effect of intensive glycemic control on macroangiopathy in NIDDM patients in the Kumamoto study was also evaluated. There were no significant differences in blood pressure levels and lipid profiles between the MIT and CIT groups. Levels of serum cholesterol, triglycerides and HDL cholesterol in both the MIT and CIT groups did not significantly change during the 6-year period. At the initiation of the study, no patient in either the MIT or CIT groups showed abnormal findings in ECG examination. In the MIT group, during the 6-year follow-up period, 1 patient died suddenly, probably due to myocardial infarction, and another patient complained of

intermittent claudication. In the CIT group, 1 patient died of cerebral infarction, another patient developed angina pectoris and 2 patients complained of intermittent claudication. The total number of major cardiovascular, cerebrovascular and peripheral vascular events in the CIT group was twice that of the MIT group (1.3 vs. 0.6 events per 100 patient-years). It has been well known that atherosclerosis occurs earlier in diabetic patients than in non-diabetic patients [27,28]. Fu et al. [29] reported that the relative risk of the development of heart disease significantly increased 1.17-fold for every 1% of increase of HbA_1c. Although this might be open to discussion because of the rather smaller number of patients in the Kumamoto study, intensive glycemic control seems to have a beneficial effect on the progression of macrovascular complications.

It has been reported that intensive insulin treatment can induce side-effects, such as hypoglycemia or weight gain. DCCT showed that the risk of severe hypoglycemia was 3-fold higher with intensive insulin therapy than that with conventional insulin therapy [30]. However, in our study, 6 patients in the MIT group and 4 patients in the CIT group had 1 or more episodes of milder hypoglycemic reactions. This might be due to the difference in insulin dosage between our study for NIDDM and DCCT for IDDM. Dosage of insulin in the MIT group was about 0.4 U/kg of body weight/24 h for NIDDM patients in the Kumamoto Study, while the dosage was 0.7 U/kg of body weight/24 h for IDDM patients in DCCT. During the 6-year follow-up study in our Kumamoto study, all patients were assigned to intensive care — visiting an out-patient diabetes clinic every 2 weeks and adjusting insulin dosage according to their degrees of glycemic control. There was a slight but not significant increase in BMI in both the MIT and CIT groups from the baseline to 6 years (MIT group, 20.5 ± 2.1 to 21.2 ± 2.1 kg/m^2; CIT group, 20.3 ± 2.8 to 21.9 ± 2.8 kg/m^2). This increase was also smaller than that reported from DCCT [2,31] and the Stockholm study [9].

Clinically, it is important to indicate the glycemic threshold to prevent the development and progression of diabetic microvascular complications. No worsening of retinopathy and nephropathy was observed in NIDDM patients whose HbA_1c, FBG and 2-h post-prandial blood glucose concentration were below 6.5%, 110 mg/dl and 180 mg/dl, respectively. Therefore, the glycemic threshold to prevent the onset and progression of diabetic microangiopathy is indicated as follows; $HbA_1c < 6.5\%$, FBG < 110 mg/dl, and 2-h post-prandial blood glucose concentration < 180 mg/dl.

The present randomized prospective study with a limited number of NIDDM patients was performed in the Department of Metabolic Medicine, Kumamoto University School of Medicine, collaborated with 2 University-affiliated hospitals. A complex multicenter, randomized study supported by Diabetes Mellitus Research Grants, the Ministry of Health and Welfare (Japan), is now in progress to draw the final conclusion on the relationship between diabetes control and complications in NIDDM patients. As stated in the report from DCCT [2], intensive insulin therapy in our study was also successfully carried out by an expert team of diabetologists, nurses and dietitians, and the time, effort and cost required were considerable. Exercise therapists and behavioral specialists did not participate in our study.

In conclusion, intensive glycemic control can delay the onset and the progression of the early stages of diabetic retinopathy, nephropathy and neuropathy in NIDDM patients as well as in IDDM patients. It remains, however, to solve the goal of glycemic control or glycemic threshold in patients with advanced microvascular complications, since it has been known that intensive glycemic control deteriorates the advanced stage of diabetic microvascular complications, such as pre-proliferative or proliferative retinopathy and nephropathy showing an albumin excretion of >300 mg/24 h [32–37].

Acknowledgements

This study was supported in part by Diabetes Mellitus Research Grants, the Ministry of Health and Welfare, Japan. The authors are grateful to Dr. H. Maruyama, Department of Cardiovascular Disease, Minamata General Medical Center, Kumamoto, Dr. S. Ueno, Department of Metabol-

ic Medicine, Arao Citizen's Hospital, Kumamoto, and many other staff for their assistance throughout the follow-up study.

References

[1] The DCCT Research Group (1986) The Diabetes Control and Complications Trial (DCCT): Design and methodologic considerations for the feasibility phase. Diabetes 35, 530–545.

[2] The Diabetes Control and Complications Trial Research Group (1993) The effect of intensive treatment of diabetes on the development and progression of long-term complications in insulin-dependent diabetes mellitus. N. Engl. J. Med. 329, 977–986.

[3] Feldt-Rasmussen, B., Mathiesen, E.R., Jensen, T., Lauritzen, T. and Deckert, T. (1991) Effect of improved metabolic control on loss of kidney function in Type I (insulin-dependent) diabetic patients: an update of the Steno studies. Diabetologia 34, 164–170.

[4] Feldt-Rasmussen, B., Mathiesen, E.R. and Deckert, T. (1986) Effect of two years of strict metabolic control on progression of incipient nephropathy in insulin-dependent diabetes. Lancet II, 1300–1304.

[5] The Kroc Collaborative Study Group (1988) Diabetic retinopathy after two years of intensified insulin treatment. J. Am. Med. Assoc. 260, 37–41.

[6] Brinchmann-Hansen, O., Dahl-Jørgensen, K., Sandvik, L. and Hanssen, K.F. (1992) Blood glucose concentrations and progression of diabetic retinopathy: the seven year results of the Oslo study. Br. Med. J. 304, 19–22.

[7] Dahl-Jørgensen, K., Hanssen, K.F., Kierulf, P., Bjøro, T., Sandvik, L. and Aagenaes, Ø. (1988) Reduction of urinary albumin excretion after 4 years of continuous subcutaneous insulin infusion in insulin-dependent diabetes mellitus: the Oslo study. Acta Endocrinologica (Copenhagen) 117, 19–25.

[8] Holman, R.R., Mayon-White, V., Orde-Peckar, C. et al. (1983) Prevention of deterioration of renal and sensory nerve function by more intensive management of insulin-dependent diabetic patients. A two-year randomized prospective study. Lancet I, 204–208.

[9] Reichard, P., Berglund, B., Britz, A., Cars, I., Nilsson, B.-Y. and Rosenqvist, U. (1991) Intensified conventional insulin treatment retards the microvascular complications of insulin-dependent diabetes mellitus (IDDM): the Stockholm Diabetes Intervention Study (SDIS) after 5 years. J. Int. Med. 230, 101–108.

[10] Reichard, P., Nilsson, B.-Y. and Rosenqvist, U. (1993) The effect of long-term intensified insulin treatment on the development of microvascular complications of diabetes mellitus. N. Engl. J. Med. 329, 304–309.

[11] Kishikawa, H., Hashimoto, Y., Hashiguchi, Y. et al. (1991) Insulin treatment regimes for Type II (non-insulin dependent) diabetes mellitus as judged by residual B-cell function. Diabetes Nutr. Metab. Clin. Exp. 4, 305–312.

[12] Diabetes Mellitus, Report of a WHO Study Group (1985) Technical Report Series 727, WHO, Geneva.

[13] Early Treatment Diabetic Retinopathy Study Research Group (1980) Early Treatment Diabetic Retinopathy Study. Manual of Operations. Bethesda, MD: Public Health Service.

[14] Early Treatment Diabetic Retinopathy Study Research Group (1991) ETDRS report No. 10: Grading diabetic retinopathy from stereoscopic color fundus photographs-an extension of the modified Airlie House classification. Ophthalmology 98, 786–806.

[15] Early Treatment Diabetic Retinopathy Study Research Group (1991) ETDRS report No. 12: Fundus photographic risk factors for progression of diabetic retinopathy. Ophthalmology 98, 823–833.

[16] Mogensen, C.E. (1988) Management of diabetic renal involvement and disease. Lancet II, 673–689.

[17] Lee, E.T. (1980) Statistical methods for survival data analysis. Lifetime Learning Publications, Belmont, CA., pp. 88–92, 127–129, 306–312.

[18] Lauritzen, T., Frost-Larsen, K., Larsen, H.W., Deckert, T. and The Steno Study Group (1983) Effect of 1 year of near-normal blood glucose levels on retinopathy in insulin-dependent diabeties. Lancet I, 200–204.

[19] Dahl-Jørgensen, K., Brinchmann-Hansen, O., Hanssen, K.F., Sandvik, L., Aagenaes, Ø. and Aker Diabetes Group (1985) Rapid tightening of blood glucose control leads to transient deterioration of retinopathy in insulin dependent diabetes mellitus: the Oslo study. Br. Med. J. 290, 811–815.

[20] Shigeta, Y., Haneda, M. and Kikkawa, R. (1992) Early detection and progression of nephropathy in patients with non-insulin-dependent diabetes. In: Y. Goto, T.H. Lee, T.Kaneko and Y. Shigeta (Ed.), Prevention and treatment of NIDDM. Smith-Gordon, London, pp. 289–291.

[21] Kuzuya, K., Akanuma, Y., Akazawa, Y. and Uehata, T. (1994) Prevalence of chronic complications in Japanese diabetic patients. Diabetes Res. Clin. Pract. 24 (Suppl.), S159–S164.

[22] Parving, H.-H., Andersen, A.R., Smidt, U.M., Hommel, E., Mathiesen, E.R. and Svendsen, P.A. (1987) Effect of antihypertensive treatment on kidney function in diabetic nephropathy. Br. Med. J. 294, 1443–1447.

[23] Mathiesen, E.R., Borch-Johnsen, K., Jensen, D.V. and Deckert, T. (1989) Improved survival in patients with diabetic nephropathy. Diabetologia 32, 884–886.

[24] Parving, H.-H. and Homel, E. (1989) Prognosis in diabetic nephropathy. Br. Med. J. 299, 230–233.

[25] Sima, A.A.F. and Brismar, T. (1985) Reversible diabetic nerve dysfunction: Structural correlates to electrophysiological abnormalities. Ann. Neurol. 18, 21–29.

[26] Sima, A.A.F., Nathaniel, V., Bril, V., McEwen, T.A.J. and Green, D.A. (1988) Histopathological heterogeneity of neuropathy in insulin-dependent and non-insulin-dependent diabetes, and demonstration of axo-glial dysjunction in human diabetic neuropathy. J. Clin. Invest. 81, 349–364.

[27] Rosengren, A., Welin, L., Tsipogianni, A. and Wilhelmsen, L. (1989) Impact of cardiovascular risk factors on coronary heart disease and mortality among middle aged diabetic men: a general population study. Br. Med. J. 299, 1127–1131.

[28] Manson, J. E., Colditz, G.A. and Stampfer, M.J. et al. (1991) A prospective study of maturity-onset diabetes mellitus and risk of coronary heart disease and stroke in women. Arch. Int. Med. 151, 1141–1147.

[29] Fu, C.-C., Chang, C.-J., Tseng, C.-H. et al. (1993) Development of macrovascular diseases in NIDDM patients in Northern Taiwan — a 4-year follow-up study. Diabetes Care 16, 137–143.

[30] The DCCT Research Group (1991) Epidemiology of severe hypoglycemia in the diabetes control and complications trial. Am. J. Med. 90, 450–459.

[31] The Diabetes Control and Complications Trial Research Group (1988) Weight gain associated with intensive therapy in the diabetes control and complications trial. Diabetes Care 11, 567–573.

[32] Lawson, P.M., Champion, M.C., Canny, C. et al. (1982) Continuous subcutaneous insulin infusion (CSII) does not prevent progression of proliferative and preproliferative retinopathy. Br. J. Ophthalmol. 66, 762–766.

[33] Puklin, J.E., Tamborlane, W.V., Felig, P., Genel, M. and Sherwin, R.S. (1982) Influence of long-term insulin infusion pump treatment of type I diabetes on diabetic retinopathy. Ophthalmology 89, 735–747.

[34] Ballegooie, E.V., Hooymans, J.M.M., Timmerman, Z. et al. (1984) Rapid deterioration of diabetic retinopathy during treatment with continuous subcutaneous insulin infusion. Diabetes Care 7, 236–242.

[35] Ohkubo, Y., Kishikawa, H., Miyata, T. et al. (1994) Optimal glycemic control for diabetic patients with advanced retinopathy. In: G. Mimura, R.-L. Qian. and K. Murakami (Eds.), Current status of diabetes mellitus in East Asia, Elsevier, Amsterdam, pp. 289–292.

[36] Ellis, D., Avner, E.D., Transue, D., Yunis, E.J., Drash, A.L. and Becker, D.J. (1983) Diabetic nephropathy in adolescence: Appearance during improved glycemic control. Pediatrics 71, 824–829.

[37] Yoshimizu, H., Nunoi, K., Nomiyama, R. et al. (1990) Persistent proteinuria developed after strict control of hyperglycemia in three cases of non-insulin dependent diabetes mellitus. In: T.H. Lee, K.B. Huh and S. Baba (Eds.), Recent advances in insulin therapy, Elsevier, Amsterdam, pp. 221–224.